

### 5.3.9.2 PSI oscillator characteristics

**Table 38. PSI oscillator characteristics**

| Symbol                                  | Parameter                                               | Conditions                               | Min <sup>(1)</sup> | Typ                    | Max                       | Unit |
|-----------------------------------------|---------------------------------------------------------|------------------------------------------|--------------------|------------------------|---------------------------|------|
| f <sub>PSI</sub>                        | PSI potential frequency                                 | If reference is HSE                      | -                  | 100                    | -                         | MHz  |
|                                         |                                                         | at 8, 16, 24, 32 or 48 MHz               | -                  | 144                    | -                         |      |
|                                         |                                                         | or HSIDIV18                              | -                  | 160 <sup>(2)</sup>     | -                         |      |
|                                         |                                                         | If reference is HSE at 25 or 50 MHz      | -                  | 100                    | -                         |      |
|                                         |                                                         |                                          | -                  | 141.67                 | -                         |      |
|                                         |                                                         |                                          | -                  | 158.33 <sup>(2)</sup>  | -                         |      |
|                                         |                                                         | If reference is LSE at 32 KHz            | -                  | 100.008                | -                         |      |
|                                         |                                                         |                                          | -                  | 144.015                | -                         |      |
|                                         |                                                         |                                          | -                  | 160.006 <sup>(2)</sup> | -                         |      |
| DuCy(PSI) <sup>(3)</sup>                | Duty Cycle                                              | -                                        | 45                 |                        | 55                        | %    |
| t <sub>su</sub> (PSI) <sup>(4)</sup>    | PSI startup time                                        | On 32 KHz                                | -                  | 850                    | 1750                      | μs   |
|                                         | PSI startup time                                        | On 8 MHz                                 | -                  | 25                     | 45                        | μs   |
| I <sub>DD</sub> (PSI) <sup>(3)(5)</sup> | PSI supply regulation bloc oscillator power consumption | If HSI not ON                            | -                  | 91                     | -                         | μA   |
|                                         | PSI oscillator power consumption                        | 100 MHz                                  | -                  | 95                     | -                         |      |
|                                         | PSI oscillator power consumption                        | 144 MHz                                  | -                  | 68                     | -                         |      |
|                                         | PSI oscillator power consumption                        | 160 MHz                                  | -                  | 77                     | -                         |      |
| N <sub>T</sub> jitter <sup>(3)</sup>    | Next transition jitter <sup>(6)</sup>                   | On PSIDIV3<br>(48 MHz with 32 kHz CK_IN) | -                  | 46.8                   | 264                       | ps   |
|                                         |                                                         | On PSIDIV3<br>(48 MHz with 8 MHz CK_IN)  | -                  | 52.6                   | 276                       | ps   |
| P <sub>T</sub> jitter <sup>(3)</sup>    | Paired transition jitter <sup>(7)</sup>                 | On PSIDIV3<br>(48 MHz with 32 kHz CK_IN) | -                  | 54.4                   | 331                       | ps   |
|                                         |                                                         | On PSIDIV3<br>(48 MHz with 8 MHz CK_IN)  | -                  | 60.6                   | 367                       | ps   |
| P <sub>er</sub> jitter <sup>(3)</sup>   | Period Jitter standard deviation                        | On PSIS<br>(100 MHz)                     | -                  | 15.5                   | -                         | ps   |
|                                         |                                                         | On PSIDIV3<br>(33.33 MHz)                | -                  | 27                     | -                         | ps   |
|                                         |                                                         | On PSIS<br>(144 MHz)                     | -                  | 14                     | -                         | ps   |
|                                         |                                                         | On PSIDIV3<br>(48 MHz)                   | -                  | 24.5                   | -                         | ps   |
|                                         |                                                         | On PSIS<br>(160 MHz)                     | -                  | 13.5                   | -                         | ps   |
|                                         |                                                         | On PSIDIV3<br>(53.33 MHz)                | -                  | 23                     | -                         | ps   |
| LT jitter <sup>(3)</sup>                | Long term jitter ethernet                               | On PSIS<br>(100 MHz with 32 kHz CK_IN)   | -                  | -                      | 13.7 (RMS)<br>96.7 (peak) | ns   |
|                                         |                                                         | On PSIS                                  | -                  | -                      | 0.79 (RMS)                | ns   |